From Q to R

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Abstract

We will explore the core concepts of convergence, continuity, and the foundational role of real numbers in analysis. In addition, we will delve into the concept of set countability. The classes will include a range problems, suitable for both those completely new to these concepts and to those who may have seen some of the material before.

1. The boring ε stuff

Definition 1.1. A real sequence is a function $a:\mathbb{N}\to\mathbb{R}$. If a is a sequence we typically write a_n for $a(n)$. We also write the sequence a as (a_n) to emphasize it's a sequence.

Definition 1.2. A sequence (a_n) is said to *tend to a limit* l if, given any positive ε, there is a natural number N (depending on ε) such that

$$|a_n - l| < \varepsilon \text{ for all } n > N.$$

We then say (a_n) tends to l (as n tends to ∞) and we write $a_n \longrightarrow l$ or $\lim_{n\to\infty} a_n = l$.

We say a sequence (a_n) is *convergent* if there exists l such that $a_n \longrightarrow l$, if there is no such l, we say the sequence is *divergent*.

Exercise 1.1. Which of these sequences are convergent?

- $a_n = \frac{1}{n}$,
- $a_n = \frac{1}{n}$, if n is a prime and $a_n = 0$ if n is not a prime,
- $a_n = \sin(\frac{n\pi}{100})$,
- $a_n = \frac{1}{\tau(n)}$, where $\tau(n)$ is the number of positive divisors of n.

We now generalise the idea of convergence to real functions, whose domain is not necessarily $\mathbb{N}$. Let $f: D\to\mathbb{R}$ be a function with domain $D\subseteq\mathbb{R}$.

Definition 1.3. We say $f(x)$ *tends to* l, *as* x *tends to* c if, given $\varepsilon > 0$, there exists δ (again, naturally depending on ε) such that

$$|f(x) - l| < \varepsilon \text{ for all } x \text{ in } 0 < |x - c| < \delta.$$

We write $f(x) \longrightarrow l$ as $x \longrightarrow c$.

Observe that in this definition we only care about the behaviour of f around c, not at c itself.

Definition 1.4. The function f is *continuous at* c if $f(x) \longrightarrow f(c)$ as $x \longrightarrow c$.

2. Why do we need the reals?

Observation 2.1. If we restrict to rationals in the definitions above we don't run into any problems. Concepts of convergence and continuity are still well-defined.

In fact, convince yourself that the function $f:\mathbb{Q}\to\mathbb{Q}$ with $f(x)=x$ is continuous. However, when we restrict to rationals we lose all the nice/sensible properties we want continuous functions to have. Here we state some theorems that real continuous functions satisfy, and we see how they fail in rationals.

Theorem 2.2. (Extreme value theorem) Let $f:[a,b]\to\mathbb{R}$ be a continuous real function. Then f is bounded and it achieves its bounds. In other words, there exist $M,m\in[a,b]$ such that $f(m)\leqslant f(x)\leqslant f(M)$, for all $x\in[a,b]$.

Theorem 2.3. (Intermediate value theorem) Let f be the same as above. Then f achieves every value between $f(a)$ and $f(b)$. In other words, for every $d\in[f(a),f(b)]$ there exists a $d\in[a,b]$ such that $f(c)=d$.

Exercise 2.1. Construct a continuous function on rationals failing the intermediate value property.

Exercise 2.2. Construct a continuous function on rationals that is bounded on a closed interval $[0,1]$ but does not achieve its upper or lower bound.

Exercise 2.3. Construct a continuous function on rationals that is unbounded on a closed interval.

Hint: You might want to heavily exploit the following:

Theorem 2.4. There is no rational number satisfying $x^2=2$.

Moreover, observe that the set $\{x\in\mathbb{Q}:x^2<2\}$ does not have a least upper bound in rationals - there is no best rational upper bound for this set. We define reals as the *totally-ordered field* in which every bounded non-empty set has a least upper bound. For the sake of completeness:

Definition 2.5. Let $\mathbb{R}$ be a set with binary operations $+$ and $\cdot$ called *addition* and *multiplication* with a binary relation $\leqslant$. The set $\mathbb{R}$ contains two distinct elements we call 0 and 1. We then have the following axioms.

1. $\mathbb{R}$ is a field under multiplication and addition, where 0 is the additive and 1 is the multiplicative identity.
2. $\mathbb{R}$ is totally ordered under $\leqslant$.
3. For all $x,y,z\in\mathbb{R}$, if $x\leqslant y$ then $x+z\leqslant y+z$. For all $x,y\in\mathbb{R}$, if $0\leqslant x$ and $0\leqslant y$ then $0\leqslant x\cdot y$.
4. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound, i.e. if $A\subset\mathbb{R}$ has an upper bound, then it has a least upper bound s, such that for every upper bound s' of A, $s\leqslant s'$.

Corollary 2.6. The set $\mathbb{R}$ contains rationals as we know them.

Corollary 2.7. Every bounded increasing sequence of reals is convergent. The limit is given by the least upper bound.

Theorem 2.8. (Axiom of Archimedes) Given $r \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > r$.

Theorem 2.9. (Cantor's intersection property) Let $I_n = [a_n, b_n]$ be a sequence of nested closed intervals of positive length, so that $I_1 \subset I_2 \subset I_3 \subset \dots$. Then the intersection is non-empty,

$$\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset.$$

Exercise 2.4. How does this fail in rationals? Give a counter-example.

Exercise 2.5. * Give an example of a non-Archimedean (fails the Axiom of Archimedes) ordered field.

Proposition 2.10. (Decimal expansions) Let (a_n) be a "0-1" sequence (so $a_n = 0$ or $a_n = 1$ for every n). Let $s_n = \sum_{k=1}^{n} a_k 2^{-k}$. Then the sequence (s_n) converges. We write the limit as $r = \sum_{n=1}^{\infty} a_n 2^{-n}$.

Proposition 2.11. Let (a_n) and (b_n) be two disctinct "0-1" sequences which don't end in infinitely many "1"s. Then the reals given by the decimal expansions are different, ie

$$\sum_{n=1}^{\infty} a_n 2^{-n} \neq \sum_{n=1}^{\infty} b_n 2^{-n}.$$

3. Counatable VS Uncountable

Definition 3.1. Let A and B be sets and f a function $f : A \to B$. We say f is

- *injective*, if the following implication holds: $f(x) = f(y) \implies x = y$, for every $x, y \in A$.
- *surjective*, if for all $b \in B$ there exists $a \in A$ such that $f(a) = b$.
- *bijective*, if it is both injective and surjective.

Definition 3.2. We say a set S is *countable* if there exist an injection into $\mathbb{N}$, ie there is a function $f : S \to \mathbb{N}$, which is injective.

If there is no injection into $\mathbb{N}$, we say S is *uncountable*.

Proposition 3.3. The set of rationals $\mathbb{Q}$ is countable.

Proof. Write for every $q \in \mathbb{Q}$ write $q = \frac{a}{b}$ where $b \in \mathbb{N}$ and $a \in \mathbb{Z}$, with $\gcd(a, b) = 1$. Especially for $q = 0$, write $0 = \frac{0}{1}$. Then we construct a map $\mathbb{Q} \to \mathbb{N}$ as

$$\frac{a}{b} \mapsto \begin{cases} 2^a 3^b & \text{if } a > 0 \\ 2^{|a|} 3^b \cdot 5 & \text{if } a \leqslant 0 \end{cases}$$

Check that this map is indeed injective.

□

Exercise 3.1. Construct a bijection from $\mathbb{Z}$ to $\mathbb{N}$.

Exercise 3.2. Prove $\mathbb{N}\times\mathbb{N}$ is countable.

Exercise 3.3. Construct a bijection from $\mathbb{Q}$ to $\mathbb{N}$.

Proposition 3.4. The set of "0-1" sequences which don't end in infinitely many "1"'s is uncountable.

Exercise 3.4. Deduce that the power-set of naturals $\mathcal{P}(\mathbb{N})$, the set of all subsets of $\mathbb{N}$, is uncountable.

We now use the decimal expansions to deduce the following.

Theorem 3.5. The set of reals $\mathbb{R}$ is uncountable.

Theorem 3.6. A countable union of countable sets is a countable set: Let I be some countable index set, and for every $\alpha\in I$ the set A_α is countable. Then $\bigcup_{\alpha\in I} A_\alpha$ is countable.

Exercise 3.5. Give an alternative proof that the set of rationals is countable.

Theorem 3.7. (Schröder–Bernstein theorem) Let A and B be any sets. If there is an injective map $A\to B$ and an injective map $B\to A$, then there is a bijection between A and B.

Exercise 3.6. Construct a bijection from $[0,1]$ to $[0,1)$.

Exercise 3.7. Construct a continuous bijection from $\mathbb{R}$ to $(0,1)$.

4. Problems

1. Prove that we can't have an uncountable family of disjoint discs in the plane.
2. Let $\mathcal{C}$ be a collection of circles in the plane such that for each point a on the x-axis there is a circle in $\mathcal{C}$ passing through point a and tangent to the x-axis (at point a). Prove that $\mathcal{C}$ contains a pair of circles that intersect.
3. Show that $\mathbb{R}^2$ (the plane) is not the union of a countable collection of lines.
4. Let $\mathcal{F}$ be a collection of disjoint closed intervals of the real line of positive length. Prove that $\mathcal{F}$ is countable.
5. We say a function f on naturals $f:\mathbb{N}\to\mathbb{N}$ is *increasing* if for all naturals $m\leqslant n$, $f(m)\leqslant f(n)$. Similarly, if for all $m\leqslant n$, $f(m)\geqslant f(n)$ we say f is *decreasing*. Prove that the set of decreasing functions on naturals is countable and that the set of increasing functions on naturals is uncountable.
6. Let $\mathcal{S}$ be a collection of subsets of $\mathbb{N}$, such that for every $A,B\in\mathcal{S}$ we have $A\subset B$ or $B\subset A$. Can $\mathcal{S}$ be uncountable?
7. Let $\mathcal{S}$ be a collection of subsets of $\mathbb{N}$, such that for every $A,B\in\mathcal{S}$ the intersection $A\cap B$ is finite. Can $\mathcal{S}$ be uncountable?
8. Let $f:\mathbb{R}\to\mathbb{R}$ be an increasing function. Let C be the set of its discontinuities, $C:=\{c\in\mathbb{R}: f \text{ is not continuous at } c\}$. Prove that C is countable.

9. Prove that the function $f:\mathbb{R}\to\mathbb{R}$ with

$$f(x)=\begin{cases}1 & \text{if } x\in\mathbb{Q}\\ 0 & \text{if } x\notin\mathbb{Q},\end{cases}$$

is no-where continuous. Furthermore, modify this function to construct a function

- that is continuous only at one point,
- that is continuous only at irrationals.

10. Is there a continuous bijection from $[0,1]$ to $[0,1)$?

11. Let f be a continuous bijection between $[0,1]$ and some set $S\subseteq\mathbb{R}$. Prove that S is a closed interval, ie. $S=[a,b]$ for some $a,b\in\mathbb{R}$.

12. Is there a real function that is continuous only at rationals?

13. Find a bijection from $\mathbb{Q}$ to $\mathbb{Q}/\{0\}$. Find an order-preserving bijection from $\mathbb{Q}$ to $\mathbb{Q}/\{0\}$ (that is, $f(x)<f(y)$ whenever $x<y$). Can there be an order-preserving bijection from $\mathbb{R}$ to $\mathbb{R}/\{0\}$?

14. Cover the set $\mathbb{Q}\cap[0,1]$ with countably many closed intervals of positive length, whose lengths sum to <1. Now, find a covering where the sum of lengths is at most $\varepsilon>0$.

15. Can the interval $[0,1]$ be written as a disjoint union of open intervals of positive length?

16. * Is there an uncountable subset C of the reals which is no-where dense (meaning that for every $c\in C$ there is some $\varepsilon>0$ such that $[c-\varepsilon,c+\varepsilon]\cap C=\emptyset$)?